Tutorial 2

Problem 1. $A = \{1, 2, 2, 3, 3, 3, 4, 4, 4, 4\}$

$B = \{4, 3, 3, 2, 2, 2, 1, 1, 1, 1\}$

$$SIM_{bags}(A, B) \stackrel{def}{=} \frac{\sum_x \min(n_A(x), n_B(x))}{|A| + |B|}$$

$$= \frac{1 + 2 + 2 + 1}{10 + 10} = \frac{6}{20} = \frac{3}{10}$$

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Distance Measures: Problem 1

Space $X = \{0, 1, 2, 3, \ldots\}$

a) Not a distance measure: $d(x, x) = 0$, but $\max(5, 5) = 5 \neq 0$

b) Distance measure: This is a standart distance

c) Not a distance measure: $sum(5, 5) = 10 \neq 0$

Problem 2. $x = (5, 6, 7)$

$y = (8, 2, 4)$

$$d_{L_1}(x, y) = |5-8| + |6-2| + |7-4| = 3 + 4 + 3 = 10$$

$$d_{L_2}(x, y) = \sqrt{3^2 + 4^2 + 3^2} = \sqrt{9 + 16 + 9} = \sqrt{34}$$

$$d_{L_\infty}(x, y) = \max(3, 4, 3) = 4$$

Problem 3

$x = 0\,0\,0\,0\,0\,0$
$y = 1\,1\,0\,0\,1\,1$
$z = 0\,1\,0\,1\,0\,1$
$w = 0\,1\,1\,1\,0\,0$

d - is the Hamming distance.

$d(x,y) = 4$, $d(x,z) = 3$, $d(x,w) = 3$

$d(y,z) = 3$, $d(y,w) = 5$, $d(z,w) = 2$

Problem 3.

$x = (x_1, \dots, x_n) \in \mathbb{R}^n$

$y = (y_1, \dots, y_n) \in \mathbb{R}^n$

$z = (z_1, \dots, z_n) \in \mathbb{R}^n$

d is the L_∞ distance: $d(x,y) = \max_i |x_i - y_i|$

1) $d \geq 0$ 2) $d(x,x) = 0$ and if $d(x,y) = 0 \Rightarrow$ all $|x_i - y_i| \leq d(x,y) \Rightarrow x = y$

$|x_i - y_i| = 0$

3) $\max_i |x_i - y_i| = \max_i |y_i - x_i| \Rightarrow d(x,y) = d(y,x)$

4) Let us prove the triangular inequality:

$$d(x,z) + d(z,y) \geq d(x,y).$$

$d(x,y) = \max_i |x_i - y_i|$. Let i_0 be an index such that $\max_i |x_i - y_i| = |x_{i_0} - y_{i_0}|$. Then for real numbers we have:

$$|x_{i_0} - y_{i_0}| \leq |x_{i_0} - z_{i_0}| + |z_{i_0} - y_{i_0}|$$

So, $d(x,y) \le |x_{i_0} - z_{i_0}| + |z_{i_0} - y_{i_0}| \le$

$\le \max_i |x_i - z_i| + \max_i |z_i - y_i|$

$= d(x,z) + d(z,y)$

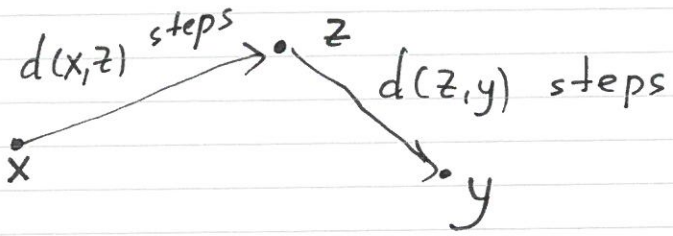
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Problem 5. 1) $d_{edit} \ge 0$

2) $d_{edit}(x,y) = 0 \iff x = y$

3) $d_{ed}(x,y) = d(y,x)$

4)



$\Rightarrow \exists$ $d(x,z) + d(z,y)$ steps from x to y.

$d(x,y)$ - smallest! $\Rightarrow d(x,y) \le d(x,z) + d(z,y)$

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kNN

Problem 1.

a) 1NN : 1-nearest to? is 5.75 $\Rightarrow$ Class = "+"

3NN : 3 nearest neighbours: 5.75 (+) ; 5.0 (−) and 4.5 (−) or 6.5 (−), So majority = Class = "−"

b) 1NN: "weigted" is the same as normal 1KNN

Answe = +

3NN : weight of + classes: $\frac{1}{(0.25)^2} = \frac{1}{1/16} = 16$ ✓

weight of − classes: $1/(0.5)^2 + 1/1^2 = 4 + 1 = 5$

] class = "+

Problem 2. $d = d_{L_\infty}$

$d(7,1) = \max(0, 0.65) = 0.65$

$d(7,2) = \max(0, 0.05) = \underline{0.05}$ (+)

$d(7,3) = \max(0.25, 0.55) = 0.55$

$d(7,4) = \max(0.25, 0.05) = \underline{0.25}$ (−)

$d(7,5) = \max(0.5, 0.3) = \underline{0.5}$ (−)

$d(7,6) = \max(0.5, 0.2) = \underline{0.5}$ (+)

Four ($K=4$) nearest are: 2, 4, 5, 6.

Weight of "+": $\frac{1}{(0.05)^2} + \frac{1}{(0.5)^2} = \frac{1}{(1/20)^2} + \frac{1}{(1/2)^2} = 404$

Weight of "−": $\frac{1}{(0.25)^2} + \frac{1}{(0.5)^2} = \frac{1}{(1/4)^2} + \frac{1}{(1/2)^2} =$

$= 16 + 4 = 20$

$404 > 20 \Rightarrow$ Answer: class(7) = "+"

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Model Evaluation. Problem 1

Predicted \ Actual	spam	not spam
spam	90	200
not spam	10	700
	↑ 90+10=100	↑ 700+200=900

$tp = 90$ $fp = 200$

$\Rightarrow$ $fn = 10$ $tn = 700$

$Pr = \frac{90}{90+200} = \frac{9}{29}$

$Re = \frac{90}{90+10} = \frac{9}{10}$

$F = \frac{2}{29/9 + 10/9} = \frac{18}{39} = \frac{6}{13}$